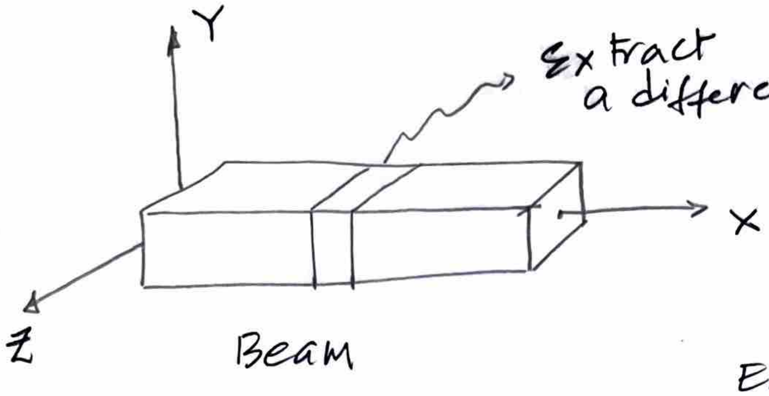
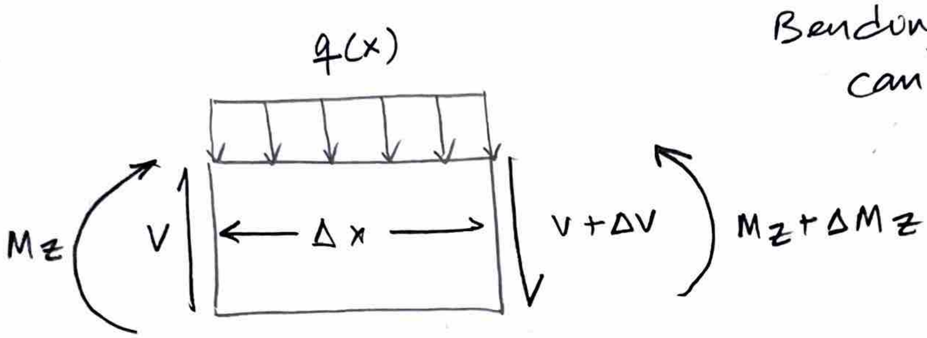
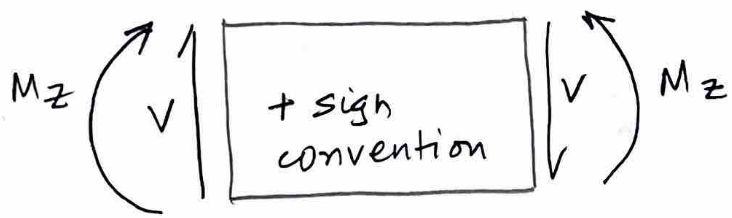
Shear forces & Bending Moments:



Extract a differential element of Length Δx

Exposed shear forces (V), loads (q) & Bending moments (M_z) can vary with X





Apply equilibrium conditions:

$$\boxed{\sum F_x = 0}$$

$$\boxed{\sum F_y = 0} \Rightarrow V - (V + \Delta V) - q(x)(\Delta x) = 0$$

$$\Rightarrow \Delta V = -q(x)(\Delta x) \Rightarrow \boxed{\frac{\Delta V}{\Delta x} = -q(x)}$$

(1)

To get a general differential equation

$$\lim_{\Delta x \to 0} \frac{\Delta V}{\Delta x} = \frac{dV}{dx} \rightarrow (2)$$

combining (1) & (2) Eqs., we have

$$\boxed{\frac{dV}{dx} = -q(x)} \rightarrow (3)$$

Assumption:

$$\int_x^{x+\Delta x} q(x)\,dx = q(x)(\Delta x)$$

{ Δx ⇝ small Length Approximation }

Apply momentum balance:

$$\boxed{\sum M_z)_A = 0}$$

$$-M_z + (M_z + \Delta M_z) - q(x)(\Delta x)\left(\frac{\Delta x}{2}\right) - (V + \Delta V)(\Delta x) = 0$$

$$\Rightarrow \Delta M_z = V(\Delta x) + (\Delta V)(\Delta x) + \frac{1}{2} q(x)(\Delta x)^2$$

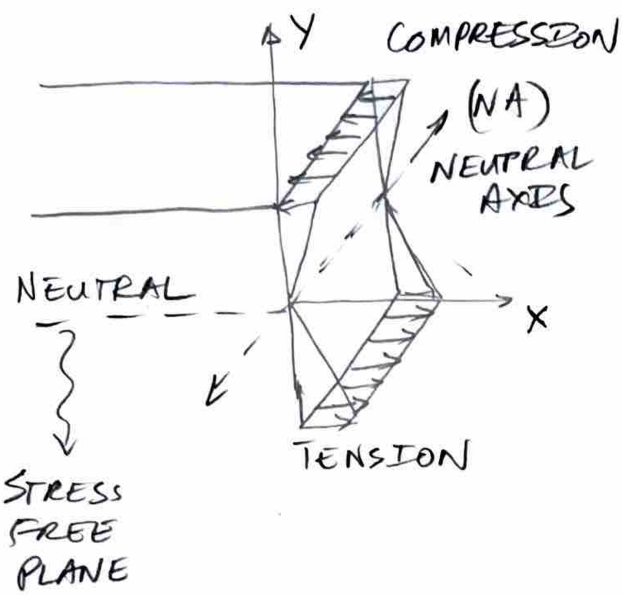
$$\Rightarrow \frac{\Delta M_z}{\Delta x} = V + \Delta V + \frac{1}{2} q(x)(\Delta x) \longrightarrow (4)$$

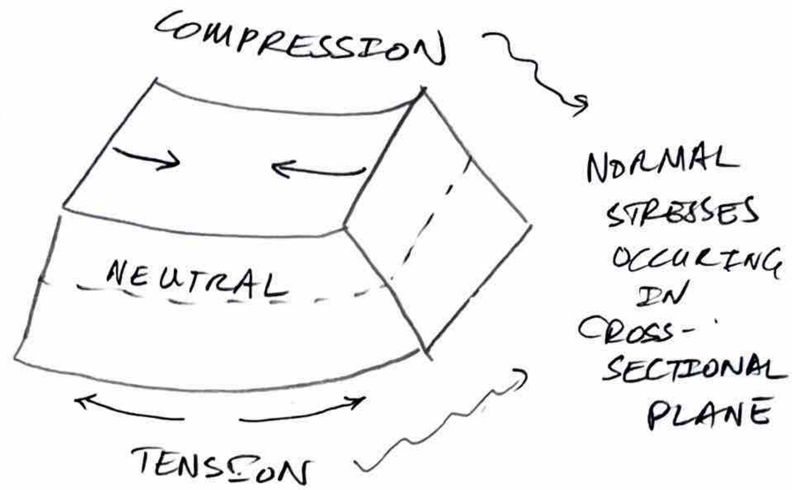
$$\lim_{\Delta x \to 0} \left(\frac{\Delta M_z}{\Delta x}\right) = \lim_{\Delta x \to 0} \left[V + \Delta V + \frac{1}{2} q(x)(\Delta x)\right]$$

$$\Rightarrow \boxed{\frac{dM_z}{dx} = V(x)} \longrightarrow (4)$$

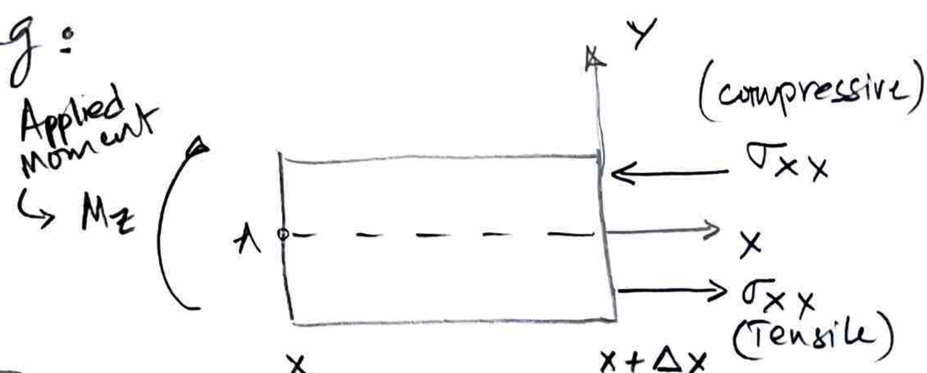
MOMENT-CURVATURE RELATION:

- Beam subjected to pure bending
- No transverse or shear loads
  - $\sigma_{xy} = 0$



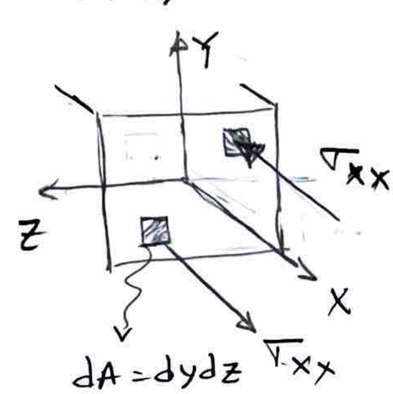


Determine normal stress in a beam subjected to pure bending:



$$\boxed{\Sigma F_x = 0}$$

$$\boxed{\int -\sigma_{xx}\, dA = 0} \longrightarrow (5)$$

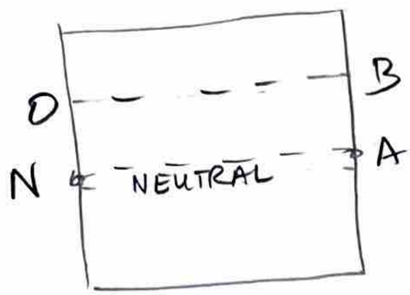


$$\boxed{\Sigma M_z = 0}$$

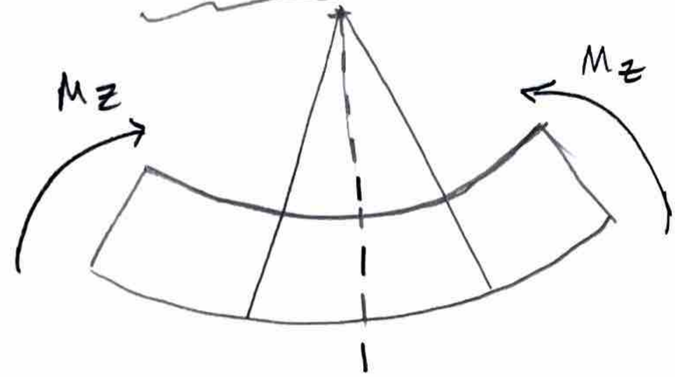
$$M_z + \int y\,\sigma_{xx}\, dA = 0$$

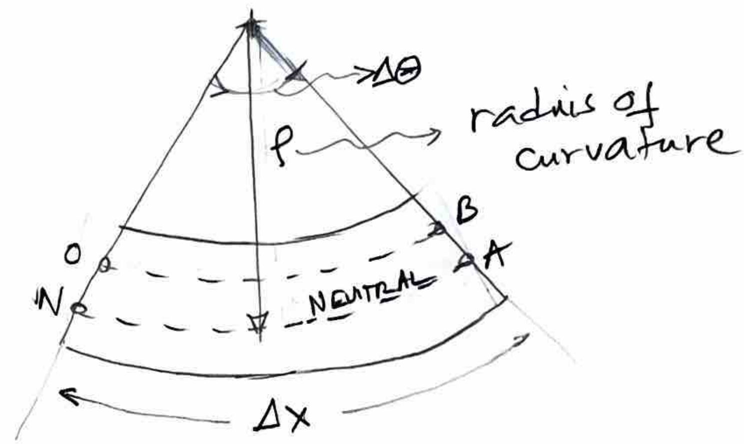
$$\Rightarrow \boxed{M_z = -\int y\,\sigma_{xx}\, dA} \longrightarrow (6)$$

Deformed differential element in pure bending:



(UNDEFORMED) XY FACE





$$\overline{NA} = \Delta x = \rho\,\Delta\theta \rightarrow (7)$$

$$\overline{OB} = \Delta\theta(\rho - y) \rightarrow (8)$$

$$\Rightarrow \overline{OB} = \rho\,\Delta\theta - y\,\Delta\theta$$

$$= \Delta x - y\frac{\Delta x}{\rho}$$

$$\overline{OB} = \Delta x\left(1 - \frac{y}{\rho}\right) \rightarrow (9) \; \{\text{from } (7) \;\&\; (8) \text{ eqs}\}$$

Recall: $\epsilon_{xx} = \left(\dfrac{\text{current length} - \text{Original length}}{\text{Original length}}\right)$

$$\Rightarrow \epsilon_{xx} = \lim_{\Delta x \to 0}\left[\frac{\Delta x\left(1 - \frac{y}{\rho}\right) - \Delta x}{\Delta x}\right] = -\frac{y}{\rho}$$

$$\Rightarrow \boxed{\epsilon_{xx} = -\frac{y}{\rho}} \rightarrow (10)$$

{Deformed radius of curvature}

5

from Hooke's Law: $\boxed{\sigma_{xx} = E\,\epsilon_{xx}}$ → (11)

(LEHI)

σ_{xx}: Normal stress; E: Youngs Modulus; ϵ_{xx}: strain

$\boxed{\sigma_{YY} = 0}$ $\boxed{\sigma_{zz} = 0}$

Isotropic behavior

From (11) & (10) Eqs:

$$\boxed{\sigma_{xx} = E\left(-\frac{y}{\rho}\right)} \rightarrow (12)$$

Recall: $\boxed{\sum F_x = 0}$ ⟹ $\int -\sigma_{xx}\, dA = 0$

Equilibrium Equations

$$\Rightarrow \int -\left(-\frac{Ey}{\rho}\right) dA = 0$$

$$\Rightarrow \frac{E}{\rho}\underbrace{\int y\,dA}_{\bar{y}A} = 0 \Rightarrow \boxed{\frac{EA}{\rho}\,\bar{y} = 0} \rightarrow (13)$$

$\bar{y}$: centroid

Explanation:

↳ E, A, ρ are non-zero constants

↳ ∴ $\bar{y} \stackrel{!}{=} 0$

⟹ z-axis must pass through centroid

$\boxed{\sum M_z = 0}$

$$\Rightarrow M_z = -\int y\,\sigma_{xx}\, dA$$

$$= -\int y\left(-\frac{Ey}{\rho}\right) dA$$

$$= \frac{E}{\rho}\underbrace{\int y^2 dA}_{I_{zz}}$$

Youngs modulus; second moment of inertia

$$\Rightarrow \boxed{M_z = \frac{E}{\rho} I_{zz}} \Longrightarrow \text{Moment-curvature relation}$$

(14)

ρ: Deformed radius of curvature

{under pure bending}

FLEXURE FORMULA:

Combine Eqs (12) & Eq (14)

$$\boxed{\sigma_{xx} = E\left(\frac{-y}{\rho}\right)} \qquad \boxed{M_z = \frac{E}{\rho} I_{zz}}$$

$$\Rightarrow \quad \boxed{\sigma_{xx} = -\frac{M_z(x)\, y}{I_{zz}}} \longrightarrow (15)$$

Normal stress at any X

General Differential Equation for Beam deflection:

$$\boxed{EI_{zz}\frac{d^2v}{dx^2} = M_z(x)}$$ (16)

vertical displacement of points along neutral axis only; $v = v(x)$

applied bending moment

E: Youngs Modulus

I_{zz}: Second moment of inertia

Second order differential equation
↳ # need two Boundary conditions

Recall relationship between shear forces & Bending moment:

$$\boxed{\frac{dM_z}{dx} = V(x)}$$ (4)

Differentiating Eq (16)

$$\frac{d}{dx}\left(EI_{zz}\frac{d^2v}{dx^2}\right) = \frac{d}{dx}M_z(x) = V(x)$$

$$\Rightarrow \boxed{EI_{zz}\frac{d^3v}{dx^3} = V(x)}$$ (17)

Beam deflection curve $v(x)$ expressed in terms of shear force $V(x)$

Relationship between Shear force and applied load:

— $V(x)$ ——— $q(x)$ —

$$\boxed{\frac{dV}{dx} = -q(x)} \longrightarrow (3)$$

Differentiating equation (17)

$$\frac{d}{dx}\left(EI_{xx}\frac{d^3v}{dx^3}\right) = \frac{dV}{dx} = -q(x)$$

$$\Rightarrow \boxed{EI_{xx}\frac{d^4v}{dx^4} = -q(x)} \longrightarrow (18)$$

↳ Beam deflection curve expressed in terms of applied load $q(x)$.